

JingZhang Co-Intelligence Belt

From a switchback railway to a co-shaped city · formal AI submission (A3 booklet)

JingZhang Co-Intelligence Belt: one belt, three cores, two axes

Design area 11.41 km2 | 294 blocks | green ratio 27.6% | public space 35.3% | key areas 192.9 + 104.3 + 72.0 ha



Cover facts

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Package path: submissions/Zhao-Yicheng/jingzhang-ai-belt
Primary / translation: zh / en (semantically equal copies)
Submitted 2026-08-31 | license COMMUNITY-DISPLAY-ONLY
Boundary status: provisional rough (official_boundary=false, recalculate later)

Statement

A conceptual reference for professional deepening; not an official conclusion, approval or commitment.
All areas recomputed from package GeoJSON in EPSG:4548; statutory controls left blank with the needed source named.

Contents and agent statement

Every deliverable is recomputable from package JSON and GeoJSON

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Agent and human review

Generated by an AI agent (Codex / OpenAI), model gpt-5-codex
Toolchain: Python, shapely, pyproj, matplotlib, all local and offline
Human direction: scope, structure, scenarios, statements and trade-offs
Data boundary: public or cleared material only, nothing internal or restricted
Review: planning, transport, utilities, heritage and participation all required
Forbidden: approved conclusions, company lists, investment or output claims
Iteration log: changelog.md records versions, adopted feedback and open items

One-glance summary

A 9.72 km spine stitches three road-severed fragments; 294 blocks tile the boundary with zero measured gap.
Layers: park band, innovation frontage, live-work clusters, then support and strategic reserve.
Green ratio 27.6% | public space 35.3% | building density 28.5% | design FAR 1.89 (concept).
Twelve scenario cards including three industry test scenarios; 5 pilgrimage landmarks with plaques.
Three phases 386.8 / 308.4 / 446.1 ha: park and transfers first, industry second, community renewal last.

Design basis and source list

formal, provisional and background material kept strictly separate

Usable as formal basis

- Announcement 1.3-1.5: purpose, three scopes, key-area areas and design tasks
- Taskbook: ten principles, three positionings, five functions, six agent tasks, boundary clause
- Urban design measures: nature and preparation of urban design deliverables
- Regulatory planning measures: statutory controls belong to the regulatory plan, not to this package
- Classification guide: authority for every land_use_code value
- Design document depth regulation: reference for deliverable depth

Provisional or background only

- Provisional rough polygons: generation, visualisation and self-check only, never a redline
- OpenStreetMap (ODbL 1.0) used only to cross-check relative positions; no derived geometry
- Case material is mechanism reference only and implies no partnership, tenancy or licence
- Repository registry and processed material are a navigation layer, not new authority

Missing data (registered, never invented)

- A-CONTROLS-001 FAR, height, density, green ratio, setbacks, redlines and control lines
- A-BOUNDARY-002 official polygons, ownership and building survey, park phases, heritage zones
- A-PROGRAM-003 / A-CLIMATE-005 capacity ratios; road sections, flows, utilities, fire and drainage baselines

prominently marked provisional: never a redline or a precise-area basis



Existing conditions and data gaps

problem list built from the announced description, never invented survey

Five structural problems

1. the corridor is repeatedly severed by roads; no continuous walkable public realm.
2. industry decoupled from housing and services; day-night imbalance and an incomplete 15-minute circle.
3. inefficient warehousing and closed compounds occupy prime frontage with unclear disposal timing.
4. station quadrants lack step-free connectivity and organised static transit; transfers break down.
5. railway, station and bell heritage are scattered without continuous narrative or identity.

gap, impact and recalculation trigger

- no official boundary: areas provisional; publication triggers full recalculation.
- no regulatory conditions: intensity conceptual until controls are published.
- no ownership survey: renewal types stay candidates until surveyed.
- no transport or utility baseline: alignments indicative pending survey and modelling.

Method, recomputable

- Continuity measured by 294 blocks and 300 m public-space accessibility (91.0%).
- Severance measured by the 9.72 km spine and crossing count; balance by jobs-housing floor area ratio.
- Flexibility measured by reserve and buffer green (%s ha) absorbing unknown disposal timing.

Overall concept and layers

one belt, three cores, two axes, many nodes; naming and case mechanisms



Structure

Belt: 9.72 km co-intelligence spine
Cores: Zhongzhiyuan, Origin Community, Dazhongsi
Axes: North 4th Ring-Xueqing, Qinghe-Xizhimen
5 landmarks plus 12 scenario nodes

Naming and identity (agent.1)

Name: JingZhang Co-Intelligence Belt (JCIB)
Logo: two rails form the character co-, green left as negative space
Palette: rail gray, brick red, compute cyan (direction only)
Fonts and marks require rights review; no licence claimed

Case mechanisms (agent.2, eight)

Kendall/MIT triple lease to whole-campus renewal | East Tech waivers to temporary scene permits
Station F and Saclay to hub and hinterland | Pangyo to a corporatised operator
one-north covenants to frontage contracts | vertical community to agent towers
lakeside corridor to ecological organisation | talent density, so scenes over subsidies

Land-use structure and areas

294 blocks tile the boundary; every class area is recomputable



Key areas, part 1

A1 the full-stack engine | A2 the first kilometre from campus to market



Key areas, part 2

A3 Dazhongsi: the public living room of agents and content

A3 Dazhongsi cluster (72.0 ha | 46 blocks)

the bell as image: soundscape landmark, bell-culture exhibition and first-store economy.
station quadrants first: parking moved out, surface returned to people.
agent vertical community: public gallery below, incubation and R&D above.
risk: interchange works unapproved; this is spatial organisation only.

key-area comparison

A1 192.9 ha | R&D led | pilot and compute | most reserve flexibility
A2 104.3 ha | housing and education | talent and prototypes | phase-one pilots
A3 72.0 ha | culture, retail and leisure | exhibition | phase-one start
sum of the three areas deviates from the announced 368.4 ha by under 0.5 per cent

public space and landmark layout

five landmarks spaced along the spine form a half-day pilgrimage and event route.
each carries a plaque: dataset, model card, evaluation and human-review record.
continuity of plazas and parks is the primary KPI, not commercial floor area.

Mobility, blue-green and public space

continuity and transfers before any commercial increment



Slow mobility and rail

9.72 km pedestrian-first spine, shuttles at defined hours
29,846 m of centreline (indicative, no redline)
three transit connectors and step-free quadrants in phase one
parking: intercept outside with P+R, underground and mechanical inside

Blue-green and public space

green 314.6 ha, ratio 27.6%, public space 402.5 ha
91.0% of area within 300 m of public space
one spine, wedges and pearls; sponge-integrated sunken greenery
barrier-free: continuous wheelchair route, handrails, seating spacing, dual guidance

Municipal and new infrastructure

one trench reserved along the spine, reducing repeat excavation; survey and fire review required.
one network: compute and distribution staged by station, district energy plus station storage.
one hub: city data hub in Zhongzhiyuan interfacing the district platform, with safety governance.

AI scenarios, personas and data governance

twelve scenario cards including three industry test scenarios

Twelve scenario cards

- SC-01 greenway continuity and break detection
- SC-02 step-free station quadrants
- SC-03 autonomous shuttle loop (test)
- SC-04 last-mile delivery and night freight (test)
- SC-05 community health assistant with GP co-sign
- SC-06 personalised learning in open classrooms
- SC-07 embodied-AI pilot shared factory (test)
- SC-08 micro-grid and storage dispatch
- SC-09 digital heritage and AI guiding
- SC-10 neighbourhood co-governance assembly
- SC-11 enterprise policy agent
- SC-12 public safety and event crowd review

Five personas (agent.3)

- 1. scientists: encounters plus cheap compute
 - 2. builders: real users and test permits
 - 3. robotics engineers: hard ground and test loop
 - 4. founders: capital and administrative certainty
 - 5. residents and elders: services with a human channel
- each card: trigger, data, carrier, service, human review, metric
scenario-space-operations mapping published in the HTML

Data governance: three hard boundaries

- 1. no identifiable personal data; face, health, residence and minors excluded entirely.
- 2. automation yields candidate advice only; disposition stays with staff, logged and appealable.
- 3. plaques publish verifiable evidence only; synthetic results may not be presented as measured.

Pilgrimage landmarks (agent.4, agent.5)

North Station plaza | Dazhongsi bell | Tsinghua Yuan memorial
herringbone test loop | global developer canopy
narrative: switchback, century of railway, co-shaped city; wayfinding by station, turnpoint and mileage

Renewal register and phasing

twelve projects, three phases, mechanisms rather than promises

Twelve demonstration projects

- 01 greenway completion and break removal
- 02 quadrant connectivity at three stations
- 03 Beizhan-Xizhimen exchange node
- 04 agent vertical community
- 05 AI Origin Memorial
- 06 embodied-AI test loop
- 07 compute and evaluation hub
- 08 warehouse to pilot factory
- 09 complete live-work community
- 10 15-minute circle gap closing
- 11 utility trench and energy station
- 12 data hub and safety platform

Phasing (386.8 / 308.4 / 446.1 ha)

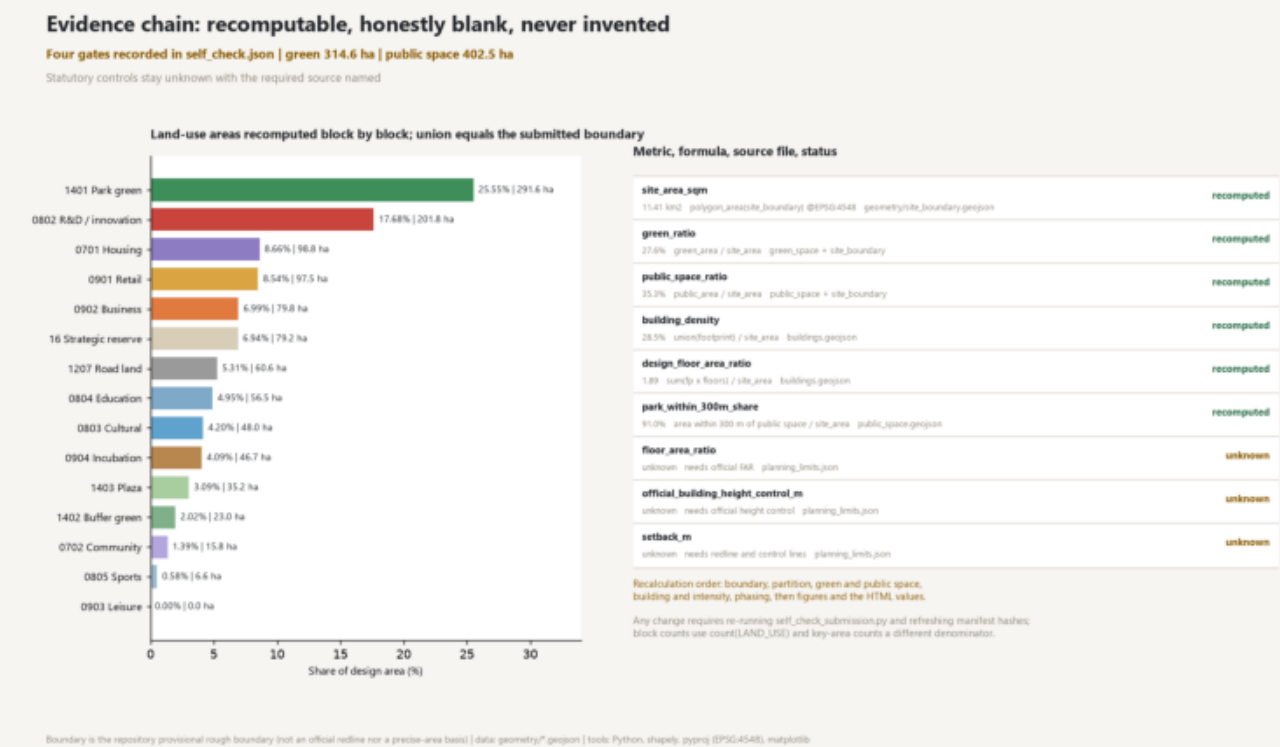
Phase 1: park continuity and two pilots (34%)
Phase 2: Zhongzhiyuan and frontage network (27%)
Phase 3: live-work renewal and reserve (39%)
basis: disposal difficulty and public-space-first
each phase end requires recalculation and redrawing
phase 1 is recomputable from PHASE-1

Implementation mechanisms and operations

- 1. opportunity list plus temporary permits make streets testable, revocable and auditable.
 - 2. authorised public data operation with exit clauses; 3. change-of-use lists and a single cross-ownership operator.
 - 4. compute and evaluation vouchers for SMEs and open-source communities; scenes instead of tax rebates.
- operations: annual assembly, quarterly open days, monthly workshops, annual plaque review.
communication: three half-day routes plus a bilingual reproducible package.

Metrics and compliance matrix

known values recomputed from geometry, unknown left blank with the source named



Compliance coverage (23 mandatory items)

Announcement 1.3.1-1.3.3, 1.4.1-1.4.3, 1.5.1.1-1.5.1.2, 1.5.2.1-1.5.2.5, 1.5.3.required-1.5.3.3
agent.1 naming, agent.2 ecosystems, agent.3 scenarios, agent.4 public space, agent.5 culture, agent.6 operations
each with nine pointer types: sections, layers, metrics, drawings, HTML modules, sources, standards, assumptions, checks

depth and standards

design depth: all 15 core items complete, limits disclosed with completeness_limited_by.
nine standards including generative-AI measures, barrier-free law and the eldercare plan.
the four gates are persisted in self_check.json after self-check.

Risk, copyright and pending data

conceptual boundaries and AI-generation disclosure

Nine risks and countermeasures

1. boundary precision: recalculate the whole chain after publication.
2. missing controls: intensity, height and setbacks marked conceptual or unknown.
3. ownership uncertain: register lists types and candidate locations only.
4. no company lists, investment, output or policy promises.
5. heritage first; special assessment before any works.
6. no redlines or sections; alignments are indicative.
7. no biometric or identifiable data; human review retained.
8. verifiable evidence required; synthetic cannot be presented as measured.
9. provisional boundaries faded and dashed, never the subject.

Copyright and toolchain

all geometry, metrics, figures, PDFs and HTML generated from site-package data (Python, shapely, pyproj, matplotlib).
no commercial tiles, remote scripts or licensed fonts; OSM background only under ODbL 1.0.
cited text summarises public information without redistribution; license COMMUNITY-DISPLAY-ONLY.

Pending data and recalculation list

official polygons, regulatory controls, redlines and sections, ownership survey, heritage lines, utilities, flow and parking baselines.
trigger: rerun build, render, finalize, self-check and preflight, then refresh every drawing.